



**CALICUT UNIVERSITY – FOUR-YEAR
UNDERGRADUATE PROGRAMME (CU-FYUGP)**

BSc PHYSICS HONOURS

Programme	B. Sc. Physics Honours				
Course Title	FOUNDATIONS OF ARTIFICIAL INTELLIGENCE				
Type of Course	Major Elective (SPECIALIZATION IV: DATA SCIENCE AND ARTIFICIAL INTELLIGENCE)				
Semester	VI				
Academic Level	300 - 399				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	4	-	-	60
Pre-requisites	Awareness of algorithmic approaches				
Course Summary	The course introduces the concept of artificial intelligence. The various knowledge representation and Knowledge Inference methods are introduced. The course introduces the application of AI in various fields.				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Able to gain insight into the evolution of key ideas and technologies by exploring the Artificial Intelligence history and its foundational concepts.	U	C	Instructor-created exams /Quiz/Assignment/ Seminar
CO2	Able to acquire knowledge and skills to understand, design, implement intelligent agents to perceive, reason and act within their environments.	U	C	Instructor-created exams/ Quiz/Assignment/ Seminar

CO3	Proficiency in various uninformed and informed search strategies along with constraint satisfaction problem solving methods.	U	C	Instructor-created exams/ Quiz/Assignment/ Seminar
CO4	Ability to design and implement logical agents and construct ontologies that capture the semantics of a domain, facilitating knowledge representation.	U	C	Instructor-created exams/ Quiz/Assignment/ Seminar
CO5	Understand the ethical considerations of AI and their societal impacts and gain insights into the future trajectory of AI by analysing the emerging trends.	U	C	Instructor-created exams/ Quiz/Assignment/Seminar
CO6	Represent various AI problems using algorithmic approaches and enhance problem-solving skills by visualizing solutions through the utilization of software tools.	U, Ap	C, P	Practical Assignment / Observation of Practical Skills
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs (48+12)	Mark
I	Introduction to AI		8	12
	1	Artificial Intelligence: Definition, Advantages and Disadvantages	1	
	2	History of Artificial Intelligence	2	
	3	Types of Artificial Intelligence	2	
	4	Applications of AI	2	
	5	The Future of AI	1	
	References Chapter 1 of Book 1, Chapter 1 of Book 2			
II	Artificial Intelligence Technologies		16	22
	6	Techniques in AI	2	
	7	Intelligence and Components of Intelligence	3	

	8	Agent and Environment	3	
	9	Informed Search Algorithms and Uninformed Search Algorithms	3	
	10	Hill Climbing Algorithm in Artificial Intelligence	3	
	11	Local Search Algorithms	2	
	References Chapter 3 of Book 1, Chapter 2 of Book 2			
III	Knowledge Representation & Reasoning		14	21
	12	Knowledge Representation , Knowledge based agents, The Wumpus world	3	
	13	Types of Knowledge ,Techniques of Knowledge Representation in AI, Logical Connectives in Propositional Logic	3	
	14	Inference Rules, Forward Chaining and Backward Chaining in AI	3	
	15	Reasoning: Probabilistic Reasoning in Artificial Intelligence	2	
	16	Bayes’ Theorem : Bayesian Belief Network in Artificial Intelligence	3	
	References Chapter 4,5 of Book 1, Chapter 12,13,14 of Book 2			
IV	Current Trends in Artificial Intelligence		10	15
	17	AI and Ethical Concerns	1	
	18	AI as a Service (AIaaS)	1	
	19	Robotics	2	
	20	Recent Trends in AI	2	
	21	Expert System: Characteristics, Components and Applications	2	
	22	Internet of Things(IoT) and Artificial Intelligence of Things (AIoT)	2	
	References Chapter 8 of Book 1, Chapter 26,27 of Book 2			
V	Open Ended Module:		12	

Books and References

1. Artificial Intelligence Beyond Classical AI by Reema Thareja , Pearson Education(Book 1)
2. Artificial Intelligence: A Modern Approach by Stuart Russell and Peter Norvig, Pearson Education (Book 2)

Mapping of COs with PSOs and POs :

	PS O1	PS O2	PS O3	PSO 4	PS O5	PS O6	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7
CO 1	2	0	0	0	1	0	3	0	1	1	2	1	0
CO 2	2	2	1	1	1	1	3	2	2	2	2	1	1
CO 3	2	2	3	1	1	1	3	1	2	2	3	1	1
CO 4	2	2	2	1	1	2	3	2	2	2	3	2	1
CO 5	1	0	1	0	1	0	3	1	1	2	3	3	1
CO 6	2	2	3	1	1	2	3	1	2	3	3	1	2

Correlation Levels:

Level	Correlation
0	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Discussion / Seminar
 - Internal Theory/Practical Exam
 - Assignments /Viva
 - End Semester Exam (70%)

Mapping of COs to Assessment Rubrics

	Internal Theory/ Practical Exam	Assignment /Viva	Practical Skill Evaluation	End Semester Examinations
CO 1	✓	✓		✓
CO 2	✓	✓		✓
CO 3	✓	✓		✓
CO 4	✓	✓		✓
CO 5	✓	✓		✓
CO 6		✓	✓	